

Document-level PII detection

Results

The pivot does not work, and the reason is worth more than the pivot would have been. On real documents with real negatives, every learned arm has a specificity of approximately **zero** — it flags essentially every document, including every genuinely PII-free one. Its document-level accuracy equals the always-say-yes baseline to three decimal places.

Measured on the two corpora that carry a document-level verdict *with* negatives — datax (prevalence 0.473) and govdocs2 (0.594), both real business and government documents, dual-judged:

Arm	F1	Precision	Recall	Specificity	Accuracy
current_rules	0.4573	0.5588	0.3869	0.6549	0.5166
hash_sgd	0.6941	0.5341	0.9994	0.0021	0.5343
champion_1k	0.6940	0.5340	0.9994	0.0015	0.5340
per_label_v4	0.6939	0.5341	0.9987	0.0026	0.5341
<i>always say yes</i>	<i>0.6934</i>	<i>0.5335</i>	<i>1.0000</i>	<i>0.0000</i>	<i>0.5335</i>
two independent judges	0.9487	—	—	—	0.9545

Read the last two rows together. The learned arms are **indistinguishable from a constant "yes"** — precision equals prevalence, recall is 1.0, specificity is

¹ Meanwhile two independent judges answer the same question at F1 **0.9487**,

Cohen's κ **0.9057**. The task is not hard to measure and not hard to do. These models simply do not do it.

current_rules — the arm dismissed at macro F2 0.0985 — is the **only** one with any discrimination: specificity 0.655, and the only accuracy above the majority-class baseline.

Why my earlier number was wrong

Scored the obvious way, across the five label_complete corpora, document-level detection looks superb: champion_1k F1 **0.9824**, accuracy 0.9676. That number is meaningless. Four of those five corpora have **100% PII prevalence** — every document is positive, so a constant "yes" scores F1 1.0:

Corpus	Prevalence	champion_1k predicted positive
ai4privacy_pii_masking_eval_10k	1.0000	10,625 / 10,626
betterdataai_ner_silver_eval_10k	1.0000	10,360 / 10,360 (all)
openpii_pii_eval_38k	1.0000	38,937 / 38,937 (all)
pii_holdout_20k	1.0000	20,000 / 20,000 (all)

Corpus	Prevalence	champion_1k predicted positive
pii2_eval_30k	0.8383	30,000 / 30,000 (all)

On pii2, the one complete corpus with genuine negatives, precision is **0.8383** — exactly its prevalence. The model flags all 30,000 documents. A 0.98 on an all-positive corpus is a measurement artifact, and reporting it as a capability would have been the "suspiciously perfect score" failure.

Root cause: the training set has almost no negatives

Split	Rows	Positive	Prevalence
Train, overall	554,247	457,421	0.8253
openpii_pii_train_155k	155,744	155,744	1.0000
ai4privacy_pii_masking_train_42k	42,504	42,504	1.0000
betterdataai_ner_silver_train_41k	41,436	41,436	1.0000
pii_trainset_100k	100,000	100,000	1.0000
pii2_train_150k	150,000	100,003	0.6667
26095_govdocs2-dualjudge-train80	26,095	13,225	0.5068
16000_datax-dualjudge-trainset	16,000	4,509	0.2818
nemotron_train_22k	22,468	0	0.0000

339,684 training documents — 61% of the set — come from corpora where every single document contains PII. Real documents run 28–51% positive, so the training mix is badly misaligned with the deployment distribution.

But "no negatives" would overstate it, and the precise version matters: **96,826 negatives do exist in the training set** (17.5%) — 49,997 in pii2, 22,468 in nemotron, 12,870 in govdocs2, 11,491 in datax. Positives outnumber them 4.7 : 1. So the models were shown negatives; they were shown 4.7× as many positives, and nothing in the objective rewarded getting the negatives right.

The recall gate is the other half of the cause

Skew alone does not force specificity to 0.0015. The threshold policy does. Every priority tag is tuned so its recall clears 0.90 on a bootstrap lower bound, on every corpus — which drives thresholds down until the model fires on almost any document. **A gate that constrains only recall selects, by construction, for a model that says yes.** That is the same mechanism as the precision collapse in the threshold report, seen at document level instead of per tag.

The corroboration is in the table above: per_label_v4, which raised every priority threshold to the gate boundary, has specificity 0.0026 against champion_1k's 0.0015 — a real improvement in the right direction, and still two orders of magnitude short of useful. And current_rules, the one arm never tuned against the recall gate at all, is the one with specificity 0.655.

Disputed documents

Both dual-judge corpora carry documents the two judges could not agree on: 1,203 in datax, 643 in govdocs2. They are excluded from the headline above and reported separately, because gold two annotators disagreed about is not gold. Every learned arm flags **100%** of them; current_rules flags 28.6% / 38.4%. That is the same finding again rather than a new one — an arm that flags everything flags the ambiguous cases too.

TL;DR

- **Document-level detection fails on real documents.** Every learned arm has specificity ≈ 0.00 and accuracy equal to the always-say-yes baseline (0.534). Precision equals prevalence. They flag everything.
- **The task itself is well-measured and very doable** — two independent judges agree at F1 **0.9487**, κ **0.9057**. This is a model failure, not a measurement failure, and it is the opposite of the tagging metric where the labels were the ceiling.
- **My earlier 0.98 was an artifact.** Four of the five label_complete corpora are 100% PII-positive; a constant "yes" scores 0.98 there. The measurement had to move to the two corpora with real negatives before it said anything.
- **Two causes, not one.** 61% of the training set comes from all-positive corpora (train prevalence 0.825 vs 0.28–0.51 in real documents) — though 96,826 negatives do exist, outnumbered 4.7 : 1. And the recall-only gate selects for a model that says yes: per label_v4, tuned to the gate boundary, has 1.7 $\times$ the specificity of champion_1k, while current_rules, never tuned against the gate, has 400 $\times$.
- current_rules **is the only arm that discriminates** (specificity 0.655). The weakest arm on the tagging metric is the only usable one on this task.
- So the pivot recommended in the previous report is **not** available off the shelf — but it is reachable, and the fix is a data problem rather than a modelling one.

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
current_rules	4000_datax-dualjudge-evalset-1.32k					
current_rules	6589_govdocs2-dualjudge-eval20-3.53k					
current_rules	ai4privacy_pii_masking_eval_10k					
current_rules	betterdataai_ner_silver_eval_10k					
current_rules	nemotron_eval_6k					
current_rules	openpii_pii_eval_38k					
current_rules	pii2_eval_30k					
current_rules	pii_holdout_20k					
hash_sgd	4000_datax-dualjudge-evalset-1.32k					
hash_sgd	6589_govdocs2-dualjudge-eval20-3.53k					

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
hash_sgd	ai4privacy_pii_masking_eval_10k					
hash_sgd	betterdataai_ner_silver_eval_10k					
hash_sgd	nemotron_eval_6k					
hash_sgd	openpii_pii_eval_38k					
hash_sgd	pii2_eval_30k					
hash_sgd	pii_holdout_20k					
hash_sgd_f2	4000_datax-dualjudge-evalset-1.32k					
hash_sgd_f2	6589_govdocs2-dualjudge-eval20-3.53k					
hash_sgd_f2	ai4privacy_pii_masking_eval_10k					
hash_sgd_f2	betterdataai_ner_silver_eval_10k					
hash_sgd_f2	nemotron_eval_6k					
hash_sgd_f2	openpii_pii_eval_38k					
hash_sgd_f2	pii2_eval_30k					
hash_sgd_f2	pii_holdout_20k					
hybrid_priority_001	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority_001	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority_001	ai4privacy_pii_masking_eval_10k					
hybrid_priority_001	betterdataai_ner_silver_eval_10k					
hybrid_priority_001	nemotron_eval_6k					
hybrid_priority_001	openpii_pii_eval_38k					
hybrid_priority_001	pii2_eval_30k					
hybrid_priority_001	pii_holdout_20k					
tfidf_linear	4000_datax-dualjudge-evalset-1.32k					
tfidf_linear	6589_govdocs2-dualjudge-eval20-3.53k					
tfidf_linear	ai4privacy_pii_masking_eval_10k					
tfidf_linear	betterdataai_ner_silver_eval_10k					
tfidf_linear	nemotron_eval_6k					
tfidf_linear	openpii_pii_eval_38k					
tfidf_linear	pii2_eval_30k					

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Docs/s ec	ms/ doc
tfidf_linear	pii_holdout_20k					
embeddingbag_asl	4000_datax-dualjudge-evalset-1.32k					
embeddingbag_asl	6589_govdocs2-dualjudge-eval20-3.53k					
embeddingbag_asl	ai4privacy_pii_masking_eval_10k					
embeddingbag_asl	betterdataai_ner_silver_eval_10k					
embeddingbag_asl	nemotron_eval_6k					
embeddingbag_asl	openpii_pii_eval_38k					
embeddingbag_asl	pii2_eval_30k					
embeddingbag_asl	pii_holdout_20k					
hybrid_priority	4000_datax-dualjudge-evalset-1.32k					
hybrid_priority	6589_govdocs2-dualjudge-eval20-3.53k					
hybrid_priority	ai4privacy_pii_masking_eval_10k					
hybrid_priority	betterdataai_ner_silver_eval_10k					
hybrid_priority	nemotron_eval_6k					
hybrid_priority	openpii_pii_eval_38k					
hybrid_priority	pii2_eval_30k					
hybrid_priority	pii_holdout_20k					
champion_1k	4000_datax-dualjudge-evalset-1.32k					
champion_1k	6589_govdocs2-dualjudge-eval20-3.53k					
champion_1k	ai4privacy_pii_masking_eval_10k					
champion_1k	betterdataai_ner_silver_eval_10k					
champion_1k	nemotron_eval_6k					
champion_1k	openpii_pii_eval_38k					
champion_1k	pii2_eval_30k					
champion_1k	pii_holdout_20k					
champion_1k_bootstrap	8 holdouts (5 complete for F2)				454.5	2.2
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				984.8	1.0
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				483.5	2.1
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				204.6	4.9

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Docs/s ec	ms/ doc
hash_sgd_readpass	8 holdouts / fixed 1,000-doc latency sample				146.2	6.8
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				674.0	1.5
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				341.6	2.9
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				175.4	5.7
hybrid_priority_001_readpass	8 holdouts / fixed 1,000-doc latency sample				130.7	7.6
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				454.5	2.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				237.5	4.2
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				147.8	6.8
hybrid_priority_readpass	8 holdouts / fixed 1,000-doc latency sample				113.3	8.8
perlabel_v4	4000_datax-dualjudge-evalset-1.32k					
perlabel_v4	6589_govdocs2-dualjudge-eval20-3.53k					
perlabel_v4	ai4privacy_pii_masking_eval_10k					
perlabel_v4	betterdataai_ner_silver_eval_10k					
perlabel_v4	nemotron_eval_6k					
perlabel_v4	openpii_pii_eval_38k					
perlabel_v4	pii2_eval_30k					
perlabel_v4	pii_holdout_20k					

Measured 2026-08-25 · n=1,000, n=126,129.